Ethical Hacking and Incident Response
CT080-3-2 (Version VE1)

Introduction and Overview

Lecturer information

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Announcement / Communication: Ms. Teams

Pre-requisites for this module

- Introduction to security and forensics technologies (CT046-3-1) or equivalent and system and networks administration (CT106-3-2)

Aims of this module

Enable students to further develop their knowledge and skills in relation to ethical hacking.

- The students will be expose to the **ethical hacking** steps including recon, identifying system vulnerabilities & system exploitation.

- Develop their knowledge on **incident handling** phases and risk mitigation plans

Course Learning outcomes, CLOs

- **At the end of this course, YOU should be able to:**

1. Explain technical concepts of ethical hacking and incident response (C2,PLO1).
2. Perform ethical hacking procedures and system exploits based on vulnerability assessment of Digital Assets using suitable tools and methodologies (A4,PLO6).
3. Propose suitable incident response plan for an organization based on the given scenario (A3,PLO9).

Mapping of CLOs with MOEs Domain

Course Learning Outcomes	Programme Learning Outcomes (PLO)												Teaching Methods	Assessment Methods
	Knowledge and Understanding	Cognitive Skills	Practical Skills	Interpersonal Skills	Communication Skills	Digital Skills	Numeracy Skills	Leadership, autonomy and responsibility	Personal Skills	Entrepreneurial Skills	Ethics and professionalism			
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11			
CLO1	√												Lecture	Final Exam
CLO2						√							Tutorial	Group Assignment - Exploitation
CLO3									√				Tutorial / Case Study	Group Assignment - Incident Response

Teaching Strategies

- Lecture
- Tutorial
- Case Study (Individual and Group)

Teaching Strategies

- ✓ The delivery method will be **2 hours lecture and 1.5 hours lab**.
- ✓ Practical tasks and tutorials will be used to re-enforce and apply theory to encourage an analytical and problem based approach to EHIR.

Teaching Strategies

- ✓ We practice on Virtual Machines that will be given in the Lab class.
- ✓ All Virtual Machines are **VMWare** Images. (**Not Virtual Box**)
  - Windows XP
  - Windows 2008
  - Windows 2000
- ✓ All the tutorials are based on the given VMs.
- ✓ We use **Kali Linux** which is a Debian-derived Linux distribution designed for **Penetration Testing**. It is maintained and funded by Offensive Security Ltd.

Assessment Method

- ✓ **An individual assignment weighted at 50%: (PLO6, PLO9).**

Section A (Practical)	Demonstration : Computer / Network Scanning, System Enumeration, System Hacking and Exploitation, Vulnerability Assessment
Section B (Report)	Incident Handling Report

- ✓ **A final exam weighted at 50%: (PLO1)**

Student Learning Time (SLT)

Course Credit Value: 3

Total Learning Hours:

- Tutorial / Case Study : 42 hours per semester
- Independent Learning Time: 60 hours

Methods of Delivery

Hence,

- We are now moving from the traditional topic based teaching to outcome-based education

Outcomes Based Education (OBE)

- OBE is education based on producing particular educational outcomes that:
  - Focus on what students can actually do after they are taught
  - Expect all learners / students to successfully achieve particular (sometimes minimum) level of knowledge and abilities

So...What is OBE?

It's

<u>NOT</u>

What we want to teach,

It's

<u>What You should learn</u>

Course Content Outline

CLO1, CLO2 & CLO3 :

Tutorial / Case Study

- Module Introduction
- Introduction to Ethical Hacking
- Information Gathering and Footprinting
- Computer and Network Scanning
- System Enumeration
- System Hacking and Exploitation
- Cryptography
- Steganography

Course Content Outline

- Introduction to incident response and digital forensics
- Digital Media Devices and Evidences
- Digital Forensics Lab and Tools
- Memory Acquisition And Analysis (Windows OS)
- Windows Live Forensics
- Understanding File System and Hard Disk
- Data Backup and Recovery

What is expected of you

- You should abide to all the rules & regulation of APU
  - Proper attire
  - No speaking of dialects
  - Attendance is compulsory and valid medical certificates or letters from parents /guardians must support any absence from class.
  - Three lateness will be equal to one absence
  - All pagers and handphones should be turned off during lectures.

What support is available for you

- Consultation hours
- Resources
  - Reference material

Essential Reading

1) Baloch, R. (2017). Ethical Hacking and Penetration Testing Guide. Florida: CRC Press.ISBN:1351381342
2) Chapple, M. and Seidl, D. (2017). CompTIA CSA+ Study Guide: Exam CS0-001. USA: John Wiley & Sons, Inc. ISBN: 1119349915
3) Årnes, A. (2017), Digital Forensics. UK: John Wiley & Sons Ltd. ISBN: 1119262402

  - Internet resources

Achievement requirements

Undergraduate:

Marks	Alphabetical Grade	Grading Point	Classification
80-100	A+	4.0	Distinction
75-79	A	3.7	
70-74	B+	3.3	Credit
65-69	B	3.0	
60-64	C+	2.7	Pass
55-59	C	2.3	
50-54	C-	2.0	
40-49	D	1.7	Fail (marginal)
30-39	F+	1.3	Fail
20-29	F	1.0	Fail
0-19	F-	0	Fail

Question and answer session

Q & A

What we will cover next

- Chap 01: Introduction to Ethical Hacking